

Total No. of Questions : 8]

SEAT No. :

PD-4854

[Total No. of Pages : 2

[6404]-385

B.E. (Artificial Intelligence and Data Science)

COMPUTATIONAL INTELLIGENCE

(417530) (2019 Pattern) (Semester - VIII)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidate:

- 1) Answer four questions from the following.
- 2) Draw neat labeled diagrams wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of non programmable electronic calculator is permitted.
- 5) Assume Suitable/Standard data if necessary.

Q1) a) Define Swarm Intelligence and Explain Ant Colony Optimization algorithm. **[8]**

- b) Write short note on: **[10]**
- i) Evolutionary Computation versus Classical Optimization
 - ii) Ant Colony Optimization algorithm

OR

Q2) a) Write short note on: **[10]**

- i) Tournament selection method
- ii) Constraint Handling

b) What are genetic operators and what is their role in evolutionary algorithms? **[8]**

Q3) a) Explain following Terminologies of Genetic Algorithm. **[10]**

- a) Search space
- b) Genes
- c) Allele
- d) Trait
- e) Genotype and Phenotype

P.T.O.

- b) Write short note on : [7]
- a) Selection operator in Genetic Algorithm.
 - b) Stopping conditions used in genetic algorithms

OR

- Q4)** a) What are types of mutation and cross over techniques? Explain in brief. [6]
- b) Explain Messy Genetic Algorithms. [6]
 - c) Explain - Binary Representations, Floating Point Representations used in Genetic Algorithms. [5]

- Q5)** a) Explain following Word embedding Techniques: [8]
- i) Bag of Words
 - ii) TF-IDF
 - iii) Word2Vec
 - iv) GloVe
- b) Explain the process of Neural Style Transfer and discuss its applications. [5]
- c) Discuss the significance of pre-trained NLP BERT models and provide examples of their applications. [5]

OR

- Q6)** a) Describe the architecture of a Neural Machine Translation (NMT) model and discuss the role of Seq2Seq in NMT. [9]
- b) Define BLEU Score and BERT Score as metrics for evaluating machine translation. Compare traditional metrics with neural metrics in machine translation evaluation. [9]
- Q7)** a) Describe the Network Theory Model in artificial immune systems. [6]
- b) Describe the working of the Artificial Immune System Algorithm. [5]
 - c) Explain the concept of danger theory in the context of artificial immune systems. [6]

OR

- Q8)** a) Explain how the Clonal Selection Theory Model contributes to the development of artificial immune systems? [6]
- b) Explain the concept of the natural immune system. Compare it with artificial immune models. [5]
 - c) Discuss the role of dendritic cells in the artificial immune system. How are dendritic cell-based models utilized in problem-solving and optimization tasks? [6]